The endpoint cut with first shared label three

Human Compression project: L044 continuation

Lemma 1 (An anchored path with at most one extra point)**.** *Let $T = \langle t_0, \dots, t_{h-1} \rangle$ be transitive on V, where $h \geq 3$, the t_i are involutions, and distant indices commute. Suppose $a \in V$ is fixed by $t_1, \dots, t_{h-1}$ and $t_{h-1} \neq 1$. If $|V| \leq h+2$, there are distinct points $a_0 = a, a_1, \dots, a_h$ such that t_i exchanges a_i, a_{i+1} and fixes $a_0, \dots, a_{i-1}$. If $|V| = h+1$, then*

$$T = \mathrm{Sym}(V), \qquad \langle t_1, \dots, t_{h-1} \rangle = \mathrm{Sym}(V \setminus \{a\}). \tag{1}$$

If $|V| = h+2$, the t_i cannot all be odd permutations of V. No intersection property or independence is assumed.

Proof. Some point in the $\langle t_0, \dots, t_{h-2} \rangle$-orbit of a is moved by t_{h-1}. Otherwise that orbit would be fixed pointwise by t_{h-1} and invariant under T, hence equal to V, forcing $t_{h-1} = 1$. Choose a shortest word w taking a to a point moved by t_{h-1}. Its point path is simple, and all earlier vertices are fixed by t_{h-1}. The image of its endpoint under t_{h-1} is a further vertex. Thus $|w| \leq |V| - 2 \leq h$.

Every index $0, \dots, h-2$ occurs in w. If index q were absent, the factors of smaller and larger index would commute. The larger factors fix a, and the smaller factors commute with t_{h-1}, which fixes a. The endpoint would consequently be fixed as well. The first occurrences of the indices are in increasing order: at the first failure, a new index at least two greater than all previous indices commutes past the whole preceding word and fixes a, so can be deleted, contradicting minimality. The word therefore has length $h-1$ or h.

In the latter case exactly one index q occurs twice. Between its two occurrences there must be a new index; otherwise they are consecutive and cancel. These intervening new indices, in increasing order, end at some index $p \geq q+1$. After the second occurrence of q, all remaining letters have index at least $p+1 \geq q+2$. Also $q \leq h-3$, because there was a new larger index between its occurrences. The second t_q therefore commutes with the remaining letters and with t_{h-1}. Deleting it preserves the property that the endpoint is moved by t_{h-1}, again contradicting minimality. Hence $w = t_0 t_1 \cdots t_{h-2}$.

This gives the displayed path, including its last t_{h-1}-edge. For $j < i$, the generator t_i commutes with every letter taking a to a_j and fixes a, so fixes a_j.

First suppose that these $h+1$ path vertices exhaust V. Descending from the last index shows

$$\langle t_i, \dots, t_{h-1} \rangle = \mathrm{Sym}(\{a_i, \dots, a_h\}). \tag{2}$$

Indeed the last mark is a single transposition. At the next step t_i has its displayed transposition and can only act further on $W = \{a_{i+2}, \dots, a_h\}$. It centralizes the already identified full symmetric group on W. Its additional restriction is trivial if $|W| \neq 2$; when $|W| = 2$ it may also be that pair's transposition, which already belongs to the later subgroup and can be removed. Thus the displayed adjacent transposition is always in the generated group, proving (2) and hence (1).

Now suppose there is one extra point z, and every t_i is odd. Descending again, every t_i is exactly its displayed path transposition. This is immediate for the last mark. At the next step the only possible additional action exchanges a_h, z, which would make t_{h-2} even and is forbidden. At any earlier step the later marks generate the full symmetric group on $W = \{a_{i+2}, \dots, a_h\}$, of size at least two. Centralization preserves this nontrivial orbit and its singleton complement

$\{z\}$ among the remaining points. The centralizer on W is trivial for $|W| \geq 3$; for $|W| = 2$ its nonidentity element would again make t_i even. Thus z is fixed by every mark, contrary to transitivity. This proves all claims. □

Retain now the full critical six-point-head hypotheses and all paid predecessors for L044: $n = 2r \geq 18$, the actual head is supported on D, and the critical tail is natural alternating. In particular the entire proof of *Pure alternating kernels in the critical prefixes* supplies the strictly growing head orbits C_k and the actual formula

$$P_k = G_{\leq k} = \{(s, u) \in \mathrm{Sym}(C_k) \times J_k : \operatorname{sgn} s = \operatorname{sgn} u\}, \tag{3}$$

with $C_3 = D \cup \{b\}$, $|C_3| = 7$, $|C_4| = 8$.

Theorem 2 (The terminal first-shared case is impossible)**.** *If colour $r-1$ has exactly one eligible edge, its first shared label cannot be* 3*.*

Proof. Assume the contrary. The two orbits of $G_{r-1} = P_{r-2}$ are $C = C_{r-2}$ and its complement A: C is its actual head orbit, and the unique crossing must connect all omission orbits. Write the unique crossing as $\{c, a\}$ with $c \in C$ and $a \in A$, and put $h = r - 4 \geq 5$. The head fixes A, and its transitive action is therefore generated by the restrictions of $\rho_3, \ldots, \rho_{r-2}$. The first of these is nontrivial on A, by the shared-label hypothesis. Every generator of index at most $r - 3$ fixes a: it commutes with ρ_{r-1} and preserves the two omission orbits, whereas the crossing is unique. Apply Lemma 1 in the reversed order

$$t_i = \rho_{r-2-i}|_A \quad (0 \leq i < h).$$

The strict prefix growth gives $|C| \geq r + 2 = h + 6$, so $|A| \leq h + 2$. The lemma also gives $|A| \geq h + 1$. There are precisely two cases.

Suppose first that $|A| = h+1$, so $|C| = h+7 = r+3$. Put $C' = C_{r-3}$ and $\Delta = C \setminus C'$. Strict growth gives $1 \leq |\Delta| \leq 2$. The group $P' = P_{r-3}$ fixes a and preserves separately $A \setminus \{a\}$ and Δ. By (1), its image on $A \setminus \{a\}$ is the full S_h. Let J' be its literal exterior image on $\Delta \sqcup A$. Its commutator subgroup is trivial on Δ, since S_2 is abelian, and maps onto $(S_h)' = A_h$ on $A \setminus \{a\}$. It follows that J' contains the actual pure alternating group on $A \setminus \{a\}$, fixing $\Delta \cup \{a\}$. These are even exterior permutations. Formula (3) therefore puts that same pure A_h in P', fixing C' as well.

The final generator ρ_{r-1} centralizes P' and hence both of its natural alternating subgroups on C' and $A \setminus \{a\}$. Their degrees are at least four, so it fixes both sets pointwise. It can move only $\Delta \cup \{a\}$, of size at most three. An even involution on at most three points is the identity, contradicting its crossing edge.

Suppose instead that $|A| = h + 2$. Then $|C| = h + 6 = r + 2$ and every strict increment from C_3 to C_{r-2} has size one. On C, each ρ_k with $3 \leq k \leq r - 2$ is a single transposition. Here are the details needed to exclude any exterior decoration on this particular orbit. For $k \geq 5$, ρ_k fixes C_{k-2} pointwise. Strict growth forces it to exchange the unique point of $C_{k-1} \setminus C_{k-2}$ with the unique point of $C_k \setminus C_{k-1}$; there are no other points in C_k on which it could act. The same assertion holds for $k = 3, 4$ by their explicit seven- and eight-point core formulas. When $k < r - 2$, the unique point of $C_{k+1} \setminus C_k$ is fixed by ρ_k, since ρ_k preserves C_k and C_{k+1}. Each subsequent new point is obtained from the previous one by a generator of index at least $k + 2$. String commutation consequently makes ρ_k fix all these later points as well. This proves the assertion on all of C.

Each ρ_k is even globally, so its restriction to A is odd. Thus every t_i in the displayed anchored action on $h + 2$ points is odd, contradicting the final assertion of Lemma 1. The two cases are exhausted. □

Attribution and boundary. The critical-head, natural-tail and prefix-kernel arguments remain complete acquired dependencies, including all induction and finite inputs. The last of these attributes the initial parity-fibre idea to the root collaborator. The shortest-word argument with one repeated label, the exact symmetric image on the exhausted path, and the use of its derived subgroup to remove at most two exterior points are the new proof here. The source route is the exceptional first-head exit in L044 Propositions 5.26 and 5.28, physical pages 34 and 36–37; the source endpoint strategy is retained in attribution. Together with the separate interior-cut proof, this excludes first shared index 3 at every unique cut of index at least 5. Other first shared indices, opposite ends, the terminal singleton and the full source contracts remain separate. No whole-original, conjecture, numerical or formalization credit is claimed.